

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (CSE) (Credit System) Degree Examinations**

January 2023

20HU504 - ENGINEERING MANAGEMENT

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full questions** choosing **Two full questions** from **Unit – I & Unit – II** each and **One full question** from **Unit – III**.

2) Use of Discrete Interest Factor Table is allowed.

Unit – I

Marks BT* CO* PO*

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|-------|---|----|-----|---|---|
| 1. a) | Define management. Explain the functions to be performed by managers to attain the set goals. | 10 | L*2 | 1 | 1 |
| b) | Pepsi Company produces a single article. Following cost data is given about its product:- Selling price per unit- Rs.40. Marginal cost per unit- Rs.24. Fixed cost per annum- Rs.16000
Calculate: (i) P/V ratio (ii) break even sales (iii) sales to earn a profit of Rs. 2,000 (iv) Profit at sales of Rs. 60,000 (v) New break even sales, if price is reduced by 10%. | 10 | L2 | 2 | 1 |
| 2. a) | What is the nature of planning? Explain the importance and purpose of planning. | 10 | L2 | 1 | 1 |
| b) | Define break-even point. Draw a break-even chart and explain its components. | 10 | L2 | 2 | 1 |
| 3. a) | Explain: i) Moving average Method. ii) Regression Method. | 10 | L2 | 1 | 1 |
| b) | Explain: i) Sole Proprietorship. ii) Partnership. | 10 | L2 | 2 | 1 |

Unit – II

4. a) An R & D project has a list of tasks to be performed whose time estimates are given in the Table.
i) Draw the project network.
ii) Find the critical path.
iii) Find the probability that the project is completed in 19 days. If the probability is less than 20%, find the probability of completing it in 24 days.

Activity	Name	T _o	T _m (in days)	T _p
1-2	A	4	6	8
1-3	B	2	3	10
1-4	C	6	8	16
2-4	D	1	2	3
3-4	E	6	7	8
3-5	F	6	7	14
4-6	G	3	5	7
4-7	H	4	11	12
5-7	I	2	4	6
6-7	J	2	9	10

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|----|--|----|----|---|---|
| b) | A company has purchased an equipment whose first cost is Rs. 1,00,000 with an estimated life of eight years. The estimated salvage value of the equipment at the end of its lifetime is Rs. 20,000. Determine the depreciation charge and book value at the end of various years using the straight line method of depreciation. | 10 | L4 | 3 | 1 |
| | | 10 | L2 | 4 | 1 |

P.T.O.

5. a) A project schedule has the following characteristics as shown in table. i) Construct PERT network. ii) Compute TE and TL for each activity. iii) Find the critical path.

Activity	Name	Time(days)	Activity	Name	Time(days)
1-2	A	4	5-6	G	4
1-3	B	1	5-7	H	8
2-4	C	1	6-8	I	1
3-4	D	1	7-8	J	2
3-5	E	6	8-10	K	5
4-9	F	5	9-10	L	7

10 L4 3 1

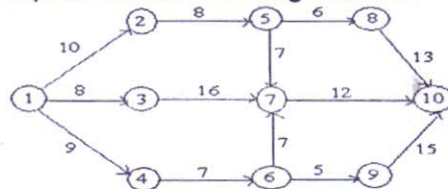
- b) The following table gives data on normal time & cost and crash time & cost of a project.

Activity	Duration (days)		Total Cost (Rs.)	
	Normal	Crash	Normal	Crash
1 - 2	3	2	300	450
2 - 3	3	3	75	75
2 - 4	5	3	200	300
2 - 5	4	4	120	120
3 - 4	4	1	100	190
4 - 6	3	2	90	130
5 - 6	3	1	60	110

Draw the network and find the critical path. If the indirect cost are Rs. 100/- per day, find out the optimum duration by crashing.

10 L3 4 1

6. a) Determine the early start and late start in respect of all node points and identify critical path for the following network.



10 L4 3 1

- b) Define Depreciation. Explain briefly the types and causes of depreciation.

10 L2 4 1

Unit – III

7. a) An amount of Rs.1200 per year is to be paid into an account each for the next five years. Using a nominal interest of 12%, determine the total amount the account will have at the end of 5th year under the following conditions. i) Deposit made at the end of each year with interest compounded monthly. ii) Deposit made at the end of each year with interest compounded continuously.
- b) A professor working in Engineering college has 10 years of service before he retires. He now plans to deposit Rs. 25,000 at the end of first year and there after an annual increase of Rs.500 for remaining nine years. If he can expect a return of 10%, find the future amount at the end of the 10th year.
8. a) Find the effective interest rate if the rate of interest is 6%, when i) Compounded yearly. ii) Compounded biannually. iii) Compounded quarterly. iv) Compounded monthly. v) Compounded daily.
- b) Discuss the interest rate from the lenders and borrowers point of view.
- c) A person deposits a sum of Rs.50,000 at an interest rate of 18% compounded annually for a period of 10 years. Find the maturity value after 10 years.

10 L4 5 1

10 L4 5 1

10 L4 5 1

5 L2 5 1

5 L4 5 1